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Physics of the Seasonal Sea Ice Zone

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Abstract

The seasonal sea ice zone encompasses the region between the winter maximum and summer minimum sea ice extent. In both the Arctic and Antarctic, the majority of the ice cover can now be classified as seasonal. Here, we review the sea ice physics that governs the evolution of seasonal sea ice in the Arctic and Antarctic, spanning sea ice growth, melt, and dynamics and including interactions with ocean surface waves as well as other coupled processes. The advent of coupled wave–ice modeling and discrete-element modeling, together with improved and expanded satellite observations and field campaigns, has yielded advances in process understanding. Many topics remain in need of further investigation, including rheologies appropriate for seasonal sea ice, wave-induced sea ice fracture, welding for sea ice freeze-up, and the distribution of snow on seasonal sea ice. Future research should aim to redress biases (such as disparities in focus between the Arctic and Antarctic and between summer and winter processes) and connect observations to modeling across spatial scales.



Sea ice extent:
the surface area of the
ocean where the sea
ice concentration
exceeds 15%

1. INTRODUCTION

The sea ice cover at both poles is now largely seasonal (**Figure 1**). According to satellite observations, the majority of Antarctic sea ice cover retreats back to the continent each year, to a degree that has been largely consistent since continuous satellite observations began in the late 1970s (**Figure 2**). On average, 15% of the maximum sea ice extent remains throughout the summer minimum, such that 85% of the Antarctic sea ice extent is seasonal. In contrast, a clear trend is evident in the Arctic: 52% of the sea ice was seasonal in the 1980s, increasing to 67% in the 2010s.

The seasonal advance and retreat of sea ice has enormous implications for weather and climate, ocean circulation, ecosystems and—in the Arctic—local communities. As sea ice retreats seasonally, it exposes darker ocean waters, lowering global albedo and enhancing the absorption of solar radiation. The newly uncovered ocean has a much greater capacity to store heat than sea ice, slowing the seasonal ocean cooling rate in autumn. Sea ice melt freshens the ocean, while sea ice growth increases its salinity, impacting ocean circulation. Sea ice provides a habitat for ice algae, a fundamental part of the food web, to remain suspended in the upper layers of the ocean and attract zooplankton. Higher-trophic-level species such as polar bears and seals rely on the ice as a physical platform, and sea ice loss has dramatically impacted these species as well as the people for whom hunting and fishing have been a traditional way of life.

Recent reviews of sea ice have focused largely on interactions between sea ice and ocean surface waves (Dumont 2022, Horvat 2022, Shen 2022, Thomson 2022) and long-term trends in sea ice (Maksym 2019). Here, we review the sea ice physics that governs the evolution of the seasonal

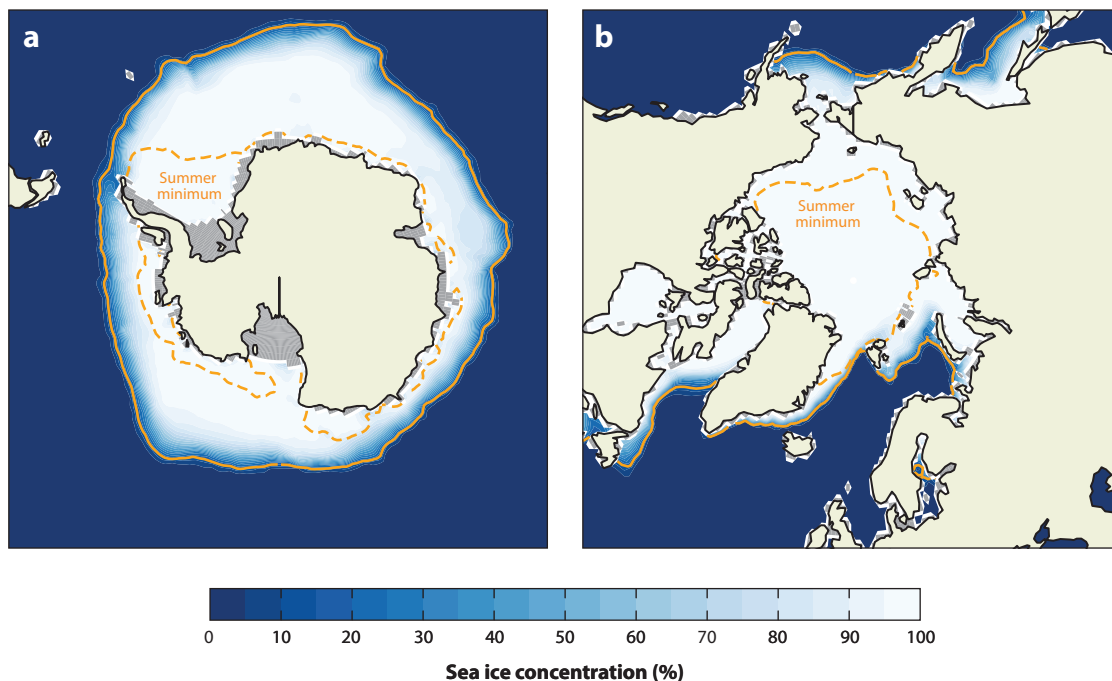


Figure 1

Mean sea ice concentrations for 2004–2023 in the months of winter maximum: (a) September in the Antarctic and (b) March in the Arctic. Orange contours show the 15% sea ice concentration contours in that month (*solid*) and the month of summer minimum (*dashed*), such that the area between the two represents the typical seasonal sea ice extent in the last two decades. Data are observations from the National Snow and Ice Data Center (NSIDC) Climate Data Record (CDR) version 4 (Meier et al. 2021).

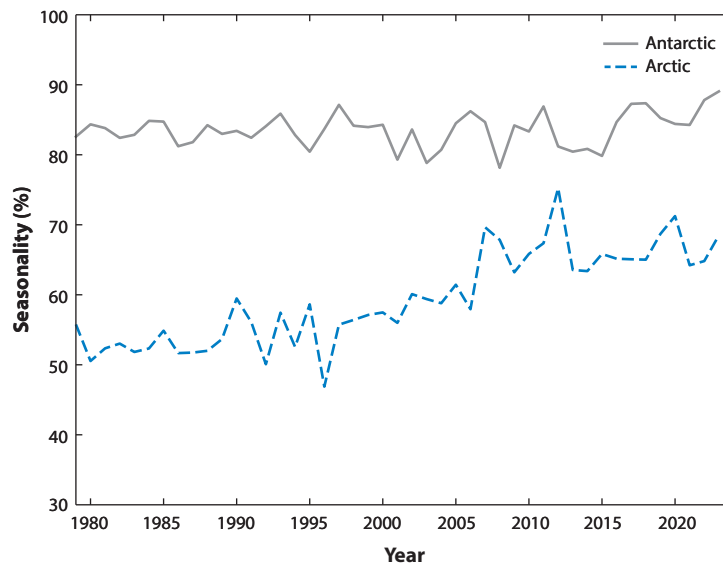


Figure 2

Sea ice seasonality based on sea ice extent (SIE), defined as $100\% \times \left(1 - \frac{\min(\text{SIE})}{\max(\text{SIE})}\right)$. Data are observations from the National Snow and Ice Data Center (NSIDC) Climate Data Record (CDR) version 4.

ice cover in both the Arctic and the Antarctic. Our review includes a broad set of processes encompassing sea ice growth, melt, and dynamics, including interactions with ocean surface waves as well as other coupled processes. We aim to provide a comprehensive view of the sea ice life cycle and to highlight processes that remain underexplored. While the physical processes involved in shaping the evolution of sea ice are intrinsically coupled with processes in the lower atmosphere and the upper ocean, this review takes a sea ice–centric perspective.

This review is structured as follows. We discuss representations of the sea ice state in Section 2 and the simple physics governing the seasonal cycle in Section 3. In Section 4, we review the current understanding of the sea ice physics relevant for seasonal sea ice, including growth, melt, and dynamics, and in Section 5 we highlight areas that require further investigation.

2. SEA ICE STATE

Seasonal sea ice is a complex, multicomponent material. Depending on local conditions and its history, a given volume of sea ice can contain a large number of individual, discrete objects (blue square in **Figure 3**), from single frazil crystals and their agglomerates up to very large floes (with length scales on the order of kilometers). The i th of these components has a unique set of N relevant properties, $p_i^1, p_i^2, \dots, p_i^N$ (density, surface and bottom roughness, thickness, diameter, elastic modulus, etc.). Even over a single floe, properties such as salinity, thickness, snow cover, and the presence of ridges vary. Together, the components form a mixture with bulk properties $P^1, P^2, \dots, P^N$ that depend on the properties of individual components in a nontrivial way. The challenge for theoretical and numerical modeling of sea ice in today's Earth system models is to find a continuum representation of this complex mixture, capturing its behavior in response to forcing from the ocean, atmosphere, and surrounding ice.

Early climate models reduced the heterogeneity of real-world sea ice cover to very simple representations, such as temperature-dependent formulations of albedo used to implicitly represent

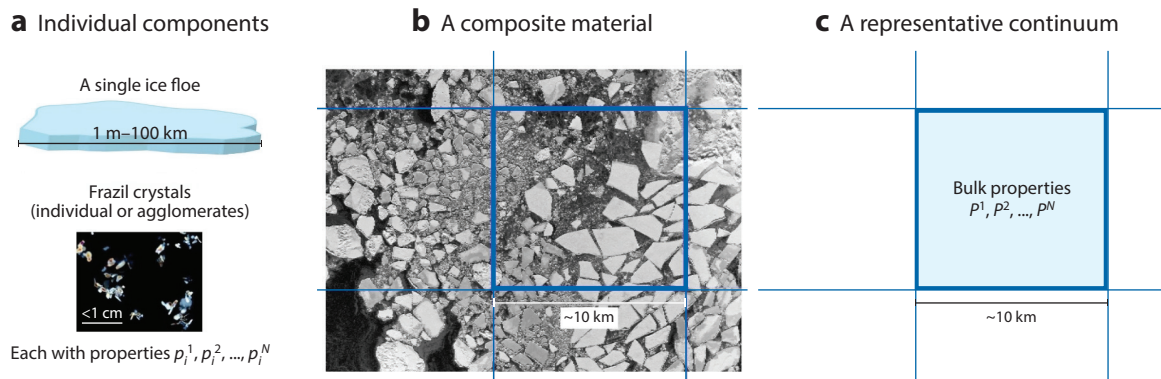


Figure 3

Sea ice, from individual components (floes, frazil crystals, etc.) to a complex mixture and its continuum representation used in models. The image in the middle shows seasonal sea ice in the northern Baltic Sea on March 19, 2020; photo by David Brus.

sea ice, fractional sea ice concentration, or grid cell mean thickness. These simple approaches were sufficient to develop understanding of the sea ice albedo feedback and polar amplification (e.g., Budyko 1969). Further detail can be gained by considering the ice thickness distribution (Thorndike et al. 1975), a probability distribution of sea ice thickness within a given area. Although seasonal sea ice is typically thinner than multiyear ice, and the thickness distribution in the Arctic has shifted over the past 30 years, ice thickness can still range from 0 to >4 m (e.g., Sumata et al. 2023), a variation that yields enormous differences in heat transfer and dynamical properties.

Another representation of sea ice is the floe size distribution (FSD), a probability function that characterizes the variability in the horizontal size of sea ice floes. Floe sizes can vary over an extremely broad range, from centimeters to hundreds of kilometers (Rothrock & Thorndike 1984). The distributions of ice thickness and floe size can be combined into a joint probability distribution (Horvat & Tziperman 2015). However, in regions where floes cannot be clearly identified or delineated, the FSD is challenging to define. This is especially true for pack ice but can also be the case in seasonal ice cover, particularly for young ice types. Even when floes are clearly separated in satellite imagery, identifying floes and determining their size distribution is not easy, which has strongly limited observations, although new methods are leading to improvements (e.g., Buckley et al. 2024). The floe size number density from satellite or aerial imagery of seasonal ice cover is often characterized by a power law, although the exponent and goodness of fit vary (e.g., Toyota et al. 2011, Stern et al. 2018, Horvat et al. 2019, Buckley et al. 2024).

In contrast to the continuum representations described above, a discrete-element representation of sea ice considers large numbers of colliding bonded elements of specified shapes and contact laws as individuals floes, which may or may not be able to change shape (e.g., Manucharyan & Montemuro 2022). Another discrete representation of sea ice is the frequency of ice type (e.g., first-year ice, grease ice, or pancake ice; see Section 4.1). Ship-based campaigns often use the Antarctic Sea Ice Processes and Climate (ASPeCt) protocol to divide sea ice into ice type, topography, thickness, and snow cover observations. It is possible to compare these categories against models (e.g., Timmermann 2004), but Earth system models typically do not consider different types of sea ice, except occasionally a separate grease ice category (e.g., Smedsrud & Martin 2015). The occurrence of different ice types is not necessarily related to a given area's ice concentration or thickness, and multiple ice types can occur within areas of a few meters, with different impacts for dynamics and heat fluxes (see Section 4).

Grease ice:

coagulation of frazil ice crystals into a soupy layer on the surface, which has a grease-like or matte appearance

Pancake ice: roughly circular pieces of ice with raised edges due to collisions with one another, typically tens of centimeters to meters in diameter

More broadly, hemispheric or regional variations in sea ice can be quantified using sea ice area, extent, or volume. Recent years have seen an increased scientific focus on the marginal ice zone (MIZ), whose extent can vary enormously depending on the definition chosen. The MIZ is defined by the World Meteorological Organization as “the region of an ice cover which is affected by waves and swell penetrating into the ice from the open ocean” (World Meteorol. Organ. 2014, p. 29), with no quantitative definition given. Based on quantities that are well observed by satellite, a quantitative definition of the MIZ that is commonly used is the area of the ice-covered ocean that is adjacent to the open ocean and where the ice concentration is less than 80% and more than 15% (e.g., Strong & Rigor 2013); however, this neither necessarily nor typically corresponds to the area of sea ice affected by waves. One could define the MIZ based on ocean surface wave properties [e.g., using the wave radiation stress (WRS); Sutherland & Dumont 2018] or FSD properties, both of which would more closely correspond with the World Meteorological Organization definition but are difficult to observe. A recently proposed definition for the Antarctic based on sea ice variability could be a new path forward (Vichi 2022). In general, which of these quantities is most useful depends on the scientific question at hand.

Here, to avoid ambiguity, we focus on the seasonal sea ice zone (SIZ) rather than the MIZ. The SIZ is the region between the winter maximum and summer minimum sea ice extent. As sea ice extent is defined by the 15% sea ice concentration contour, the SIZ has low observational uncertainty and is straightforward to quantify. The region is affected by a broad range of physics, including but not limited to those associated with ocean surface waves. It can also be thought of as the region that has been and will be an MIZ over the course of the year.

3. A FIRST-ORDER VIEW: SIMPLE PHYSICS

Seasonal changes in sea ice extent and thickness are, to first order, driven by the seasonal response to solar and longwave radiative forcing. This response can be described using a simple set of equations following, for example, Wagner & Eisenman (2015).

Consider a diffusive energy-balance model that simulates the evolution of the zonal-mean climate on an aquaplanet, neglecting the specific heat capacity of sea ice and the atmospheric column. The state variable is the enthalpy, $\mathcal{E}$, of the surface layer and the atmospheric column, which is defined as

$$\mathcal{E} \equiv \begin{cases} -L_f h, & \mathcal{E} < 0 \text{ (sea ice)}, \\ c_w (T - T_f), & \mathcal{E} \geq 0 \text{ (open water)}, \end{cases} \quad 1.$$

where L_f is the latent heat of fusion, h is the sea ice thickness, c_w is the heat capacity of the ocean mixed layer, T is the surface temperature, and T_f is the freezing point. The enthalpy, $\mathcal{E}$, evolves due to the net balance of top-of-atmosphere insolation, S , scaled by planetary coalbedo, a ; outgoing longwave radiation, which is approximated as $c_1 + c_2(T - T_f)$, where c_1 and c_2 are constants; meridional heat transport, parameterized as diffusion $D \nabla^2 T$; and a constant heat input from the ocean, F_b , such that

$$\frac{\partial \mathcal{E}}{\partial t} = a S - [c_1 + c_2 (T - T_f)] + D \nabla^2 T + F_b. \quad 2.$$

When $\mathcal{E} < 0$, Equation 2 describes the evolution of the sea ice thickness, h . In this case, the surface temperature that would balance the surface atmospheric energy flux with the vertical heat flux through ice, T_0 , is given by

$$k \frac{T_f - T_0}{h} = -a S + [c_1 + c_2 (T - T_f)] - D \nabla^2 T, \quad 3.$$

Seasonal sea ice zone (SIZ):

the extent of sea ice between the seasonal minimum and seasonal maximum extent limits

Enthalpy, $\mathcal{E}$: a state function of a thermodynamic system, defined as a sum of its internal energy and a product of its volume and pressure

Planetary coalbedo, a :

the fraction of incoming solar radiation absorbed by the Earth; in the simple model described here, it is given by a_i in the case $\mathcal{E} < 0$ (sea ice) or a_w in the case $\mathcal{E} \geq 0$ (open water), where $a_w = a_0 - a_2 \sin^2 \theta$, with θ the latitude and a_i , a_0 , and a_2 constants.

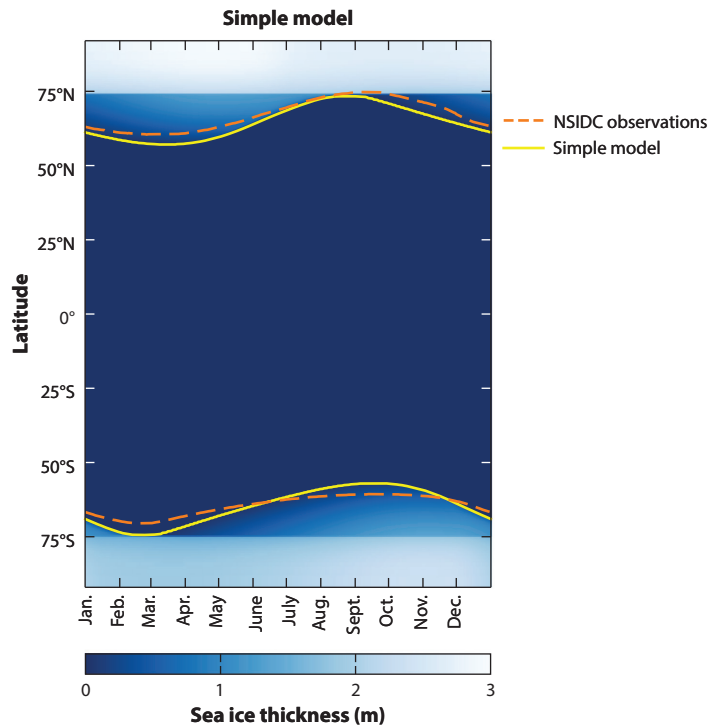


Figure 4

Zonal-mean seasonal cycle of sea ice thickness in the simple model described in Section 3. The yellow line shows the seasonal evolution of the ice edge defined by $\mathcal{E} = 0$ from the model, and the orange dashed line shows the 15% sea ice concentration contour for the 1979–1998 climatology in National Snow and Ice Data Center (NSIDC) Climate Data Record (CDR) version 4 (Meier et al. 2021).

where k is the thermal conductivity of the ice and we assume a linear temperature profile in the ice. The surface temperature is given by

$$T \equiv \begin{cases} T_f + \mathcal{E}/c_w, & \mathcal{E} \geq 0 \text{ (open water),} \\ T_f, & \mathcal{E} < 0, T_0 > T_f \text{ (melting ice),} \\ T_0, & \mathcal{E} < 0, T_0 < T_f \text{ (subfreezing ice).} \end{cases} \quad 4.$$

This model is integrated with c_1 tuned to match the observed annual-mean zonal-mean ice edge latitude and other parameters as described by Roach et al. (2022). Such a simple model can qualitatively capture the observed seasonal evolution of the zonal-mean ice edge latitude in the Arctic and Antarctic (**Figure 4**). However, this representation is an enormous simplification of the extraordinarily heterogeneous real-world sea ice cover as well as the complex air–ice–ocean dynamical and thermodynamical processes that shape it.

4. A MORE COMPREHENSIVE VIEW: THE SEA ICE LIFE CYCLE

In this section, we review the sea ice physics that governs the evolution of the seasonal ice cover in both the Arctic and the Antarctic, including sea ice growth, melt, and dynamics. The growth and melt phases can be considered a thermodynamic life cycle (**Figure 5**) that occurs each year, with the caveat that some processes may occur continuously in localized areas, such as the outer margins of the SIZ. In contrast, dynamical processes may generally occur at any time during the

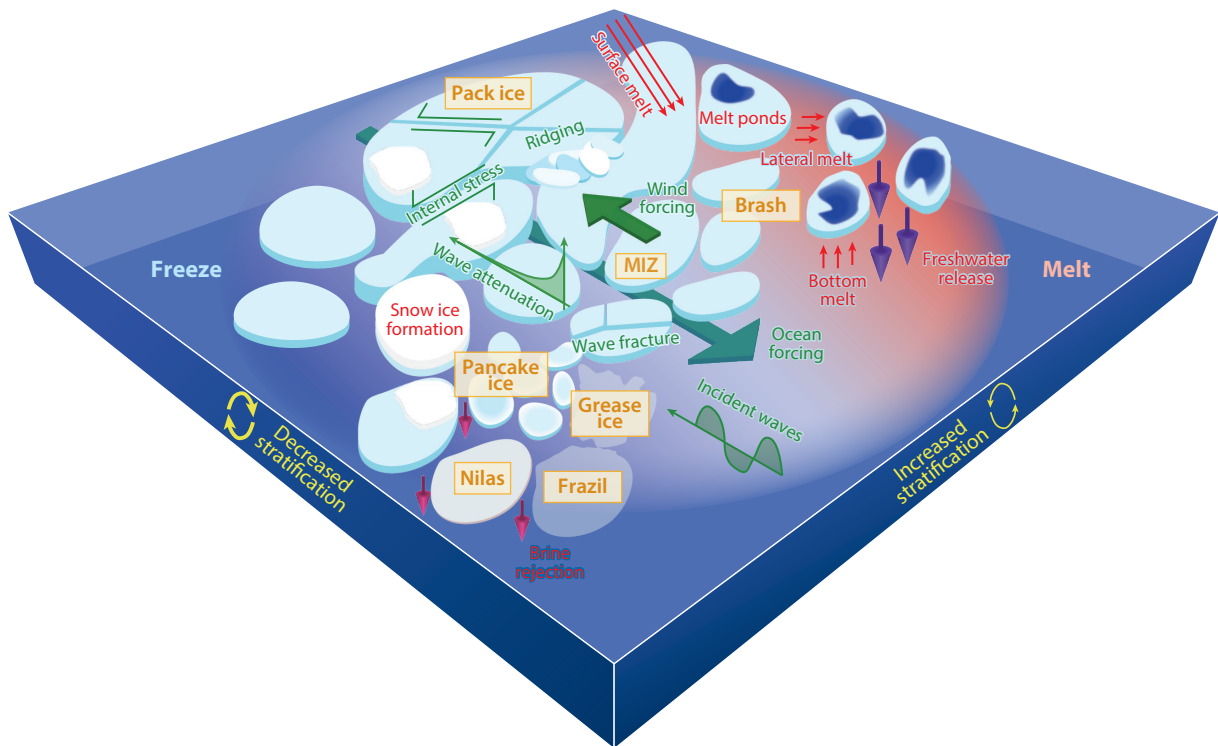


Figure 5

Growth, melt, and dynamical processes relevant for seasonal sea ice. Sea ice states are in yellow boxes, dynamical processes are in green, and thermodynamical processes are in red. Abbreviation: MIZ, marginal ice zone.

year throughout the ice cover. Their implications for sea ice mass balance and atmosphere–ocean interactions are governed by their timing within the seasonal cycle.

4.1. Sea Ice Growth

Sea ice growth occurs due to the unique properties of seawater and its interactions with the overlying atmosphere and underlying ocean. The mode of sea ice growth influences sea ice dynamics and ice–ocean–atmosphere heat transfer, playing a major role in SIZ evolution, especially during the ice advance period.

To first order, sea ice forms when the top layer of the ocean reaches the freezing point. This occurs in high latitudes, where the ocean is characterized by a permanent halocline that separates a fresher upper-ocean layer from a warm, saltier layer at depth. As seawater is cooled from above, the vertical difference in salinity limits mixing of warmer water from below, allowing the formation of sea ice (Roquet et al. 2022). The presence of salt ions lowers the freezing point for typical seawater to approximately -1.8°C . Once the upper layer has cooled below its freezing point, sea ice can begin to form on the surface of the ocean. The highest rates of new ice production occur in latent heat polynyas near the coast in both the Arctic and the Antarctic (Nakata et al. 2021), where sea ice is advected away by katabatic winds.

The properties of the sea ice that forms depend strongly on the presence and strength of ocean turbulence. This turbulence might be associated with wind shear, ocean surface waves, and/or buoyancy-driven convection (Herman et al. 2020). If the ocean surface is turbulent, freezing can

Polynya: a large, persistent region of open water and thin ice that occurs within much thicker pack ice, typically on the order of 100 km in length

Frazil ice:
millimeter-scale
needle or plate-like
crystals of sea ice

Nilas: a thin (<10 cm),
elastic layer of ice that
bends easily with
waves and, under
pressure, can raft in a
pattern of interlocking
fingers (finger rafting)

occur throughout the ocean surface layer, whereas in quiescent conditions, freezing occurs only at the surface.

Initial seed crystals in the surface ocean originate from the sublimation of water vapor in a cold atmosphere or from the freezing of water droplets produced by breaking ocean surface waves (Daly 1994). The initial crystals that form are disk shaped and typically around 0.5 mm in size (Schneck et al. 2019). If the water temperature is below freezing, crystals grow at a rate that depends on the rate at which latent heat is transported away from the crystal, which may be higher than previously thought (Rees Jones & Wells 2015). When crystals collide in the presence of turbulence, fragmented pieces become the nuclei for new crystals, a secondary nucleation process that represents a positive feedback (Clark & Doering 2009) and can be modeled as collisions between crystals belonging to discrete size classes (Holland & Feltham 2005, Heorton et al. 2017). Besides crystal growth and nucleation, the rate of gravitational removal is also an important control on frazil ice formation (Rees Jones & Wells 2018).

The dynamics of the frazil crystal and seawater mixture are driven largely by the buoyant rising of frazil crystals (Morse & Richard 2009) and turbulent diffusion. Frazil crystals precipitate into the boundary layer at the air–ocean interface, a viscous sublayer where viscous stresses dominate and the flow velocity decreases to zero at the interface. There, they can be herded by winds and/or waves into a slushy mix of seawater and frazil, termed grease ice. The grease ice accumulation rate depends on wind speed, ocean currents, drag, and resistance forces (Smedsrud 2011, Smedsrud & Martin 2015). As heat loss to the atmosphere continues, the fraction of solid ice increases. Brine is trapped between growing crystals as it forms, so new sea ice can have a salt content as much as 50% that of seawater (Wakatsuchi & Ono 1983).

If the ocean surface is turbulent, then as the grease ice consolidates, wave-induced collisions and dilation form it into near-circular floes with upturned edges, termed pancakes. The frazil-to-pancake transition takes place on a timescale of approximately 24 h (Doble 2009). The newly forming sea ice modifies the ocean surface wave field, damping wave energy, particularly at high frequencies (see Section 4.2.2). Pancake floes mostly follow wave motion (Smith & Thomson 2020). Their horizontal sizes are controlled by the properties of the ocean surface waves and are rarely less than 2 cm (Shen et al. 2001, Shen & Sankaran 2004). The maximum pancake diameter can be limited by either the tensile stress (where differential wave force creates stretching of the surface between floes) or bending stress (where vertical tensile force creates bending force on floes) according to theory developed for a monochromatic wave (Shen et al. 2001, Roach et al. 2018b).

At diameters beyond one-quarter of the wavelength, there is no longer a wave control on pancake diameter, and pancakes can grow indefinitely. Pancakes can grow in diameter via lateral growth as well as freezing together with neighbors to form composites and by incorporating interstitial frazil crystals, resulting in sheets of consolidated or welded pancakes. The rate of welding depends on the distribution of pancakes across the domain and other thermodynamic and dynamic factors (Roach et al. 2018b). The frequency and severity of pancake ice floe collisions in an active wave field increase with the wave period (Marquart et al. 2023). During collisions, pancakes may raft on top of one another; rafted pancake ice thickness strongly increases with wind speed (Dolatshah et al. 2019). These different thermodynamic and dynamic processes result in a multiregime pancake FSD (Alberello et al. 2019) and differences in pancake ice properties between the Arctic and the Antarctic (Nose et al. 2020), with the latter generally subject to stronger winds and waves.

On the other hand, if the ocean surface is calm rather than turbulent, frazil crystals may freeze together into nilas—thin, semitransparent ice sheets. Nilas is typically up to 10 cm thick and appears dark gray, and therefore has a smaller impact on albedo than pancake ice. The volume of

ice produced via pancake ice formation can be twice as high as that produced via nilas formation, as pancake ice formation allows increased heat flux from the ocean to the atmosphere during freeze-up (Doble 2009, Naumann et al. 2012).

Both pancakes and nilas subsequently thicken. The increase in thickness typically occurs via congelation growth, the downward growth of ice crystals into the water. The rate of downward congelation growth is determined by the net balance of fluxes at the ice–ocean interface and is also modestly impacted by salt transport between the two media (Wells et al. 2019). Brine loss from sea ice occurs via temperature-dependent brine pocket migration, brine expulsion, and, most importantly, gravity drainage via a network of cells and channels. Brine loss can enhance sea ice formation, as well as being a key driver of ocean circulation through the creation of dense water. In supercooled water, e.g., near Antarctic ice shelves, sea ice can also thicken due to the formation of additional frazil crystals on the underside of the existing ice (Gough et al. 2012).

An additional mode of sea ice growth is snow ice formation. If the snow accumulation (from precipitation or snow redistribution) on top of sea ice is large enough compared with the ice thickness to push the ice surface below sea level, then seawater can permeate the snow on the sea ice, which can then freeze into ice. Snow ice formation can also occur when high ocean heat fluxes lead to basal melt, sea ice thinning, and flooding of the overlying snow. Snow ice formation rates may exceed congelation growth rates regionally (Zhaka et al. 2023). Snow ice formation is more common in the Antarctic, which has high precipitation rates and thin sea ice that can more easily be depressed than it can in the Arctic, and may represent around a quarter of total sea ice formation there (Jeffries et al. 2001).

4.2. Sea Ice Dynamics

Throughout its life cycle, sea ice is subjected to a range of dynamic forcings, including winds, ocean currents, waves, and internal stresses. These forcings govern the drift of sea ice, which helps determine its extent and concentration, as well as the mechanical redistribution of sea ice via rafting and ridging, which helps determine sea ice thickness (e.g., Worby et al. 2008).

Due to its low density, sea ice is positively buoyant. Apart from frazil crystals (Section 4.1), which can move vertically in the water column depending on the balance between gravity, buoyancy, and drag, the last of which depends nonlinearly on the sizes, shapes, and orientations of crystals (e.g., McFarlane et al. 2014), all forms of ice remain at the surface and, most of the time, stay in hydrostatic equilibrium. The motion of large floes due to waves is an exception to this rule (see Section 4.2.2). Thus, in most situations, it is justified to analyze the motion and deformation of sea ice in two horizontal dimensions.

Forces acting on ice floes can be divided into two groups: those proportional to the ice mass (body forces, e.g., the along-surface component of the gravitational force in the case of nonzero gradients of sea surface height, or the Coriolis force) and those proportional to the surface area on which the force is acting (surface forces, including stresses from the ocean, atmosphere, and surrounding ice). The surface stresses can be further divided into tangential and normal wind stresses (related to skin and form drag, respectively) acting on the parts of ice floes protruding from the water, tangential and normal oceanic stresses acting on submerged parts of ice floes, and internal stress related to collisions or friction between floes (contact stress) and to turbulent-like motion of floes (kinetic stress). The common feature of all three stress sources is their dependence on the velocity of the ice relative to the air, water, and surrounding floes. Due to the nonlinear nature of these relationships, the net stresses analyzed at sufficiently large spatial and temporal scales depend not only on the mean wind, current, and ice velocities but also on their high-frequency fluctuations: wind gustiness, turbulent ocean eddies, wave motion, and floe–floe collisions (see,

Congelation growth:

a bottom-freezing process that occurs via the downward growth of ice crystals into the water; congelation ice has a distinctive columnar crystal texture

Skin drag:

drag caused by friction between a fluid and object moving through it, influenced by the object's surface roughness and fluid viscosity

Form drag:

drag caused by the pressure difference between the front and rear surfaces of a moving object, which is influenced by the object's shape

Rheology: the study of properties of materials that determine their reaction to deformation and flow, i.e., relationships between external forces acting on a material and its internal reaction (change of shape)

Free drift: a mode of sea ice motion under equilibrium between the oceanic, atmospheric, and Coriolis forces, with zero internal stress

e.g., Feltham 2005, Andreas et al. 2010, Manucharyan & Thompson 2017, Gupta & Thompson 2022, Brenner et al. 2023, Gupta et al. 2024, and references therein). On hourly timescales, tidal currents and inertial oscillations are an additional source of stress (e.g., Womack et al. 2024).

In contrast to a dense ice pack, in which the primary force balance is between the average wind, ocean current, and internal ice stress, in seasonal ice the net effects of fluctuations of the forcing and the ice itself are significant. We are only just beginning to understand and appreciate the role of these processes. The following subsections present current knowledge on horizontal surface stresses from the wind and ocean; wave-induced processes, including vertical and horizontal stresses and the fracturing of ice floes; and internal stresses and sea ice rheology.

4.2.1. Wind and ocean stresses. The primary driver of sea ice dynamics in the SIZ is wind stress, while ocean currents typically act as a drag on sea ice motion. The stresses exerted on sea ice by wind and currents depend on ice concentration (A) and morphology, i.e., the sizes of and distances between roughness elements at the upper and lower surfaces of the ice. Depending on the ice type and state, the roughness elements can include floe edges, ridges, rafted pancakes, snow formations, subice frazil crystals, and so on. All of these features affect the bulk drag coefficients of the top and bottom surfaces of the ice ($C_{d,ai}$ and $C_{d,oi}$, respectively). Due to the large variability of surface properties, both $C_{d,ai}$ and $C_{d,oi}$ vary strongly, from well below 10^{-3} for grease ice to more than 10^{-2} for deformed ice (Brenner et al. 2021 and references therein). Importantly, in the ice pack the bulk ice–ocean and ice–atmosphere drags are highest in the range of middle ice concentrations: If A is very low (below 0.2–0.3), then the bulk drag is low because the roughness of the open water is lower than that of the ice, whereas if the floes are densely packed ($A \sim 1$), then the neighboring roughness elements are less effective due to a sheltering effect (e.g., Lu et al. 2011, Tsamados et al. 2014).

For individual ice floes, the skin drag acting on their top and bottom surfaces is proportional to their surface area, and the form drag acting on their edges is proportional to their diameter. The proportionality of surface forcing leads to a size-dependent response of floes to the forcing and is one of the mechanisms of floe clustering (Herman 2011, 2012), with patches of densely packed floes even when $A \ll 1$. Generally, however, as long as A remains below ~ 0.8 , internal stress in the ice (see Section 4.2.3) remains low and the ice is in free drift, moving at approximately 2–4% of the wind speed (Wagner et al. 2022). In the Antarctic MIZ, however, free drift is observed even at $A \sim 1$ (Womack et al. 2022). The strong influence of ice type and morphology on the drag coefficients leads to a pronounced seasonal cycle of ice–ocean and ice–atmosphere stresses observed in the Arctic, with a minimum in late autumn and a maximum in late spring (e.g., Brenner et al. 2021, Mchedlishvili et al. 2023). Relatively simple parameterizations of drag in which the drag coefficient is linearly proportional to the ice thickness, combined with a free-drift assumption, successfully capture seasonal variability in Arctic ice drift at large scales (Brunette et al. 2022).

4.2.2. Wave stresses and fracture. Ocean surface waves dynamically impact sea ice cover by influencing sea ice formation (Section 4.1) and exerting stresses that act horizontally and vertically and may cause sea ice fracture. These are coupled interactions, as sea ice in turn influences waves by attenuating wave energy. Wave–sea ice feedbacks play an important role in the evolution of the SIZ. For example, larger WRS may compact the SIZ so that it is more resistant to other stresses, while a fractured sea ice field leads to less wave attenuation, enhanced ice melt, and reduced sea ice cover. As several reviews of wave dynamics in the MIZ have recently been published (Dumont 2022, Shen 2022, Thomson 2022), we include only a brief summary here.

Swell grows in open water under wind forcing and propagates into sea ice from tens to thousands of kilometers away. Waves can also be generated locally by wind in the open ocean near sea ice or in partially ice-covered regions. Once waves enter sea ice, wave energy is attenuated with

distance into the sea ice. The decrease of wave energy, E , as a function of frequency, f , with distance into sea ice, x , is typically modeled with an exponential decay (e.g., Squire 2018, Thomson 2022):

$$E(f, x) = E(f, 0) \exp^{-\alpha(f)x}. \quad 5.$$

The attenuation rate $\alpha(f)$ increases with the frequency of the waves and is typically observed to be a power law in incident wave frequency $\alpha \propto f^p$ with $2 \leq p \leq 4$, meaning that high-frequency waves are most rapidly attenuated. Higher-order or linear attenuation rates may be more appropriate in some situations (Kohout et al. 2014, Squire 2018). The attenuation rate is also expected to depend on the state of the sea ice cover, including the ice type, thickness, concentration, and floe size. For example, it can increase fivefold in the winter compared with the spring due to changes in sea ice state (Wahlgren et al. 2023).

Waves are attenuated in sea ice by scattering or dissipation. Scattering conserves wave energy and can cause changes in wave direction and is most relevant for floes with length scales similar to the wavelength. Dissipation is nonconservative and is beginning to be thought of as the more important of the two mechanisms (Thomson 2022). The main processes by which wave energy dissipation occurs are turbulence at the ice–ocean interface arising from differential velocity between sea ice and orbital wave motion (Voermans et al. 2019), fracture of sea ice floes (Ardhuin et al. 2020), and inelastic collisions between floes (Smith & Thomson 2020). In situ observations suggest that dissipation by turbulence may be more important than floe collisions (Smith & Thomson 2020).

The dissipation of wave energy at the ice–ocean interface creates stress in the direction of the wave propagation (i.e., WRS) and can be expressed as (Longuet-Higgins & Stewart 1964)

$$\tau = -\nabla \cdot \mathbf{R} = -\frac{\partial R_{ij}}{\partial x_i} \hat{\mathbf{e}}_j, \quad 6.$$

where $\mathbf{R}$ (and its components R_{ij}) is a tensor that depends on the wave amplitude, group and phase speed and orientation, and wavenumber spectrum. By assuming that all the energy dissipated in the attenuation is transferred to the ice as momentum and that the wave is aligned in the x direction, we can simplify the equation as (Dumont 2022)

$$\tau_x = -\frac{1}{2} \rho g \frac{dE}{dx}, \quad 7.$$

where $\frac{dE}{dx}$ is given by Equation 5. The WRS pressure has been measured in the 0.1–1-Pa range and acts as a stabilizing factor for sea ice (Stopa et al. 2018). WRS typically acts from open ice into the sea ice cover and therefore compacts sea ice, making it more resistant to wind stress and deformation (Boutin et al. 2020). It can be stronger than the wind stress in low-wind conditions within approximately 50 km of the ice edge (Stopa et al. 2018).

Sea ice, at the floe scale, is a brittle material and does not deform much before breaking into several pieces. Sea ice fracture events due to waves are common at large scales (Prinsenbergh & Peterson 2011, Kohout et al. 2014) and even at the scale of a single ice floe (Herman et al. 2021). Different criteria governing the fracture of sea ice floes have been proposed (Dumont et al. 2011). The two main methods are based on (a) the deformation of the ice floe when assuming that the floe exactly follows the sea surface height field and fractures when a critical flexural strain limit is reached, and (b) the stress applied by the wave on the ice when assuming that the ice is rigid, does not deform, and fractures when the flexural strength is reached. These two approaches can be used in combination by applying the former to short waves and the latter to long waves (Dumont et al. 2011). Both are imperfect assumptions: Ice can deform due to the waves and does not exactly flex with the sea surface height field. The value for the fracture threshold based on flexural strain,

Constitutive equation: a relationship between applied forces and geometrical effects induced by them

Strain, ϵ : a nondimensional measure of the relative deformation of a body compared with its reference shape and position

Strain rate, $\dot{\epsilon}$: the time derivative of strain, measured in s^{-1}

Polydispersity: a measure of the heterogeneity of the grain sizes in a granular material

Restitution coefficient: the ratio of the relative velocity of separation after a collision to the relative velocity of approach before the collision

$I_{br} = 0.014$, agrees well across observations and laboratory experiments (Voermans et al. 2020) as well as across values found by statistical analysis in a pan-Arctic model.

Wave-induced fracture can rapidly modify the sea ice FSD. Sea ice fractures preferentially into sizes close to half the incident wavelength, giving log-normal, Gaussian, or modal FSDs according to detailed strain-based modeling applied to each floe (Mokus & Montiel 2022) or observations (Herman et al. 2021, Dumas-Lefebvre & Dumont 2023). Wave-induced fracture of sea ice exposes the edges of floes to the open ocean and can enhance lateral melt (see Section 4.3) or growth.

4.2.3. Sea ice internal stresses. Sea ice internal stress is variable and poorly understood but plays an important role in deformation. The response of a given volume of sea ice mixture to forces from the surrounding ice is described by a rheological model, consisting of constitutive equations relating the applied stress σ_{ij} to strain ϵ_{ij} and/or strain rate $\dot{\epsilon}_{ij}$ and to bulk material properties (elastic modulus, viscosity, etc.; see, e.g., Irgens 2014).

In the case of sea ice, with its wide range of sizes of crystals and/or floes, its rheological properties are closely related to the scale at which deformation is analyzed. All rheology models of sea ice assume a clear separation of scales between the characteristic size of individual floes, $\bar{d}$, and the scale L at which macroscopic deformation takes place: $\bar{d} \ll L$. Equivalently, the number of crystals or floes in the analyzed volume of ice should be sufficiently large to capture natural variability. This assumption is unproblematic in young ice types (grease ice, shuga, pancakes, and small ice floes) as well as, most of the time, at scales of tens of kilometers in an ice pack, i.e., scales corresponding to typical mesh resolution of global and regional models. In that case, the actual value of $\bar{d}$ becomes irrelevant. Accordingly, no information about floe size is used in the rheologies suitable for large-scale analysis of sea ice dynamics, including the classical viscous–plastic rheology of Hibler (1979) and its later variants (for an overview, see Feltham 2008), as well as the recently developed elasto-brittle models (Dansereau et al. 2016, Ólason et al. 2022). In Arctic-wide simulations, elasto-brittle rheology captures essential features of sea ice deformation, including many (but not all) statistical properties of the localization and scale invariance of deformation (Bouchat et al. 2022). However, for many situations in seasonal sea ice, $\bar{d}$ is both hard to determine and comparable to L : The ice is a mixture of floes spanning a very wide range of sizes, from less than a meter to tens of kilometers, and the largest floes have sizes similar to the width of the MIZ, the dimensions of a strait or channel through which ice is moving, or the sizes of ocean eddies naturally defining (and limiting) the length scale L . In that case, floe size effects cannot be disregarded, and the ice behaves as a polydisperse granular material (Herman 2022).

The only granular rheology that has been proposed for sea ice is that of Feltham (2005), based on work by Shen et al. (1984, 1986, 1987). This rheology is based on the kinetic theory of dry granular gases and thus shares important assumptions with that theory, many of which are unrealistic when applied to sea ice. In particular, it assumes that ice has a high restitution coefficient and that ice floes randomly distributed on the sea surface undergo short-lived, binary collisions but no prolonged contacts (in both cases, observations show otherwise; see Yulmetov et al. 2016, Li & Lubbad 2018, and references therein). It also disregards the presence of ice or a water–frazil mixture between floes, which can influence floes’ motion between collisions, and the nature of the collisions themselves. Moreover, although collisional rheology has been useful in some applications (e.g., Rynders et al. 2022), it can be treated as an acceptable approximation of real sea ice only under a very narrow range of conditions: The ice concentration must be sufficiently low to allow individual motion of ice floes and sufficiently high to sustain collisions between neighboring floes. A much more common situation in the SIZ, especially its inner parts, is that corresponding to a dense granular flow, i.e., a state in which a typical floe undergoes several contacts with its neighbors at the same time, and many of those contacts are long-lived.

One of the most widely used rheological models in applied research on granular materials is the so-called $\mu(I)$ rheology (Jop et al. 2006, Jop 2015). Based on simple dimensional analysis, it states that the friction coefficient μ of a granular material is a function of a single nondimensional parameter, the inertial number I , defined as $I = \dot{\epsilon}d/\sqrt{p/\rho}$, where p and ρ denote pressure and density, respectively, and d is a length scale associated with microscopic rearrangements of grains. Although the original model has been formulated for two-dimensional, dry materials in the so-called inertial regime (fast macroscopic deformation), its extended versions have been successfully fitted to observations for a wide range of materials (Kamrin et al. 2024). Thus, it proved versatile, and it is very likely that it can be applied to seasonal sea ice as well (Herman 2022). This has important consequences, because, independently of its particular version, the model has a number of features that are qualitatively different from the other rheological models of sea ice. Crucially, in the viscous–plastic and collisional rheologies, the friction coefficient μ is a constant, whereas in the $\mu(I)$ rheology, it is an increasing function of I and thus of the shear rate $\dot{\epsilon}$.

In general, recent Antarctic observations suggest that the relationship between ice concentration and internal stress is more complex than typically assumed. On the one hand, as already noted earlier, free drift in the MIZ has been observed at A close to 1 (Womack et al. 2022). On the other hand, decoupling between wind and sea ice motion observed in the MIZ in spring indicates a large role of internal stresses (Womack et al. 2024). This ice type- and season-dependent response to forcing is not yet captured by available rheological models.

4.3. Sea Ice Melt

Sea ice thermodynamic mass loss occurs fundamentally through exchanges with the atmosphere above (e.g., solar heat) or the ocean below. While this mass loss is typically referred to as melting (as we primarily do here), ablation at ice–ocean interfaces technically includes both melting and dissolution. The salinity-dependent freezing temperature and its magnitude relative to the ocean temperature determine the temperature at which mass loss occurs through melting or dissolution (Malyarenko et al. 2020). Melting is controlled predominantly by a heat transfer, which typically occurs when there is a liquid layer of similar salinity to the ice adjacent to the interface (a freshwater lens), while dissolution is controlled by a salt transfer. Dissolution generally occurs when the temperature of the ocean water is less than the freezing temperature of the ice, and the flux of salt toward the ice lowers the freezing point. The typical description of ablation using the three-equation balance requires the conservation of heat and salt throughout these processes (Holland & Jenkins 1999). The temporal variability of melt rates is governed by various processes driving the interaction of sea ice with the atmosphere above and the ocean below.

The key to understanding sea ice surface melt is albedo. The albedo (reflectivity) of the sea ice surface determines how much solar energy is absorbed by or transmitted through sea ice versus reflected back to the atmosphere (Perovich & Polashenski 2012). Going into summer, sea ice is typically snow covered (albedo 0.7–0.9; Light et al. 2022). As Arctic sea ice melts, it can develop a patchwork of bare (white) ice (albedo 0.6–0.7; Smith et al. 2022) and melt ponds (albedo 0.1–0.4; Light et al. 2022). The formation of surface melt ponds on the sea ice plays a large role in increasing the rate of melt by decreasing the albedo. Relatedly, melt ponds can drive a surface melt feedback, where surface melt contributes to expansion and deepening of ponds, which enhances melting. Episodic flushing and drainage of melt ponds often significantly raise the albedo (e.g., Polashenski et al. 2012, Webster et al. 2022). Snow plays a significant role in the interannual variability of sea ice surface melt (e.g., Perovich & Richter-Menge 2015), especially as it corresponds to the date of melt initiation or onset (Stroeve et al. 2014).

The dramatically different albedos of dominant surface types drive regional variability in the amount of surface melt. Whereas melt ponds are typically extensive on summer sea ice in the

$\mu(I)$ rheology: a rheological model that states that the friction coefficient μ of a material is a function only of the inertial number

Inertial number I : a nondimensional number relating the macroscopic timescale for bulk deformation to the microscopic timescale for grain rearrangements, computed as $I = \dot{\epsilon}d/\sqrt{p/\rho}$.

Three-equation balance: the balance among equations that determine the freezing point dependence on salinity and pressure, the conservation of heat, and the conservation of salt, which collectively govern thermodynamics at the ice–ocean interface

Arctic, resulting in high amounts of surface melt (Perovich & Richter-Menge 2015), Antarctic sea ice has very few ponds and typically thick snow cover, resulting in low ice surface melt (Drinkwater & Liu 2000). The combination of thicker snow and lower relative humidity has led to the general assumption that sea ice surface melt in the Antarctic is negligible, especially as it is typically outpaced by bottom melt.

Melt at the base of sea ice (basal melt) is driven primarily by heat in the ocean below, but the mechanisms by which this heat reaches the bottom of the ice are not always straightforward. During the melt season, relatively cold and fresh meltwater layers can buffer sea ice from warmer and saltier waters in the ocean below (Supply et al. 2022). Thus, basal ablation may actually be dominated by dissolution early in the season (Malyarenko et al. 2020), with melting taking over later in the season after ocean temperatures are warmer and storms mix up warmer waters from below (e.g., Meyer et al. 2017). Warmer waters can be brought to the surface through wind- and wave-induced turbulence (Smith et al. 2018) and small-scale ocean processes. For example, small-scale oceanic currents and surface winds accelerate the basal ice melt rate by allowing warm waters to move into colder regions under the ice floes (Gupta & Thompson 2022), and eddy-induced pumping resulting from friction between the ocean and sea ice can pull up warmer waters from below into contact with the ice (Gupta et al. 2020). A key hemispheric difference is that the Southern Ocean has relatively abundant energy for mixing and weak stratification, such that ocean heat can often be more readily accessed for melting throughout the year, including when active growth otherwise dominates in winter (e.g., Martinson 1990).

Floes undergo lateral melting at their edges driven by ocean heat in leads or surrounding open water. Waves can cause undercutting (even in relatively small leads; e.g., Richter-Menge et al. 2001), which can further accelerate the erosion of floe edges. The amount of lateral melt is determined by the perimeter of floes exposed to the ocean, which depends on the FSD. Lateral melt is likely a significant contribution to total melt when floe sizes are on the scale of meters (e.g., Bateson et al. 2020). However, as lateral and basal melt generally pull from the same ocean heat reservoir, higher lateral melt near the ice edge may be compensated for by reduced basal ice melt (Tsamados et al. 2015). The distinction is still important to the evolution of the SIZ, as the ice-albedo feedback is most active when ice melts laterally, thereby forming open water (Smith et al. 2022). Vertical melt through surface and basal melt can form open water only once ice is thin, whereas open water can form from lateral melting at any ice thickness.

5. OUTLOOK

Understanding of the processes shaping seasonal sea ice evolution has been developed from a combination of in situ, laboratory, and satellite observations and numerical modeling spanning many spatiotemporal scales (e.g., **Figure 6**). In this section, we discuss the knowledge gaps that remain, many of which can be linked to observational or modeling challenges. The SIZ is difficult to observe, due to the difficulty of making field observations from fragmented or weak ice, detecting thin and low-concentration sea ice from satellites, characterizing its small-scale heterogeneity, and capturing the episodic nature of storms and waves, among other factors. Observational limitations as well as the difficulty in representing the varied processes and sea ice states limit numerical modeling. Below, we summarize some key biases and opportunities associated with improving process-based understanding of seasonal sea ice.

5.1. Biases and Missing Processes

Observational challenges have led to a seasonal bias, with fall and winter processes associated with sea ice growth (Section 4.1) being less well understood than those associated with spring

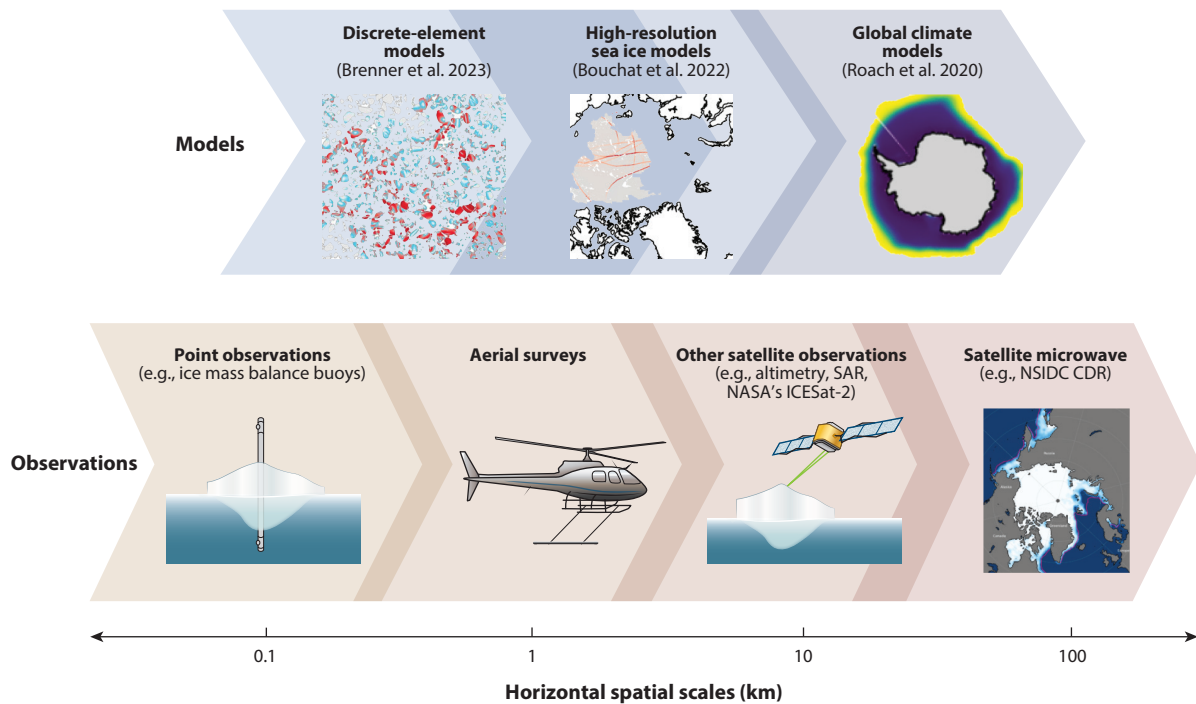


Figure 6

The spatial scales that characterize the modeling (*top*) and observational tools (*bottom*) used to understand processes driving seasonal sea ice evolution. The examples shown include (*top*) discrete-element modeling of floe velocities (image adapted with permission from Brenner et al. 2023, CC BY-NC 4.0), high-resolution modeling of sea ice deformation (image adapted from Bouchat et al. 2022, CC BY 4.0), and global climate modeling of sea ice concentration (image adapted from Roach et al. 2020), as well as (*bottom*) point measurements such as ice mass balance buoys, aerial surveys, satellite altimetry [e.g., Ice, Cloud, and Land Elevation Satellite 2 (ICESat-2)], and satellite passive microwave observations of sea ice concentration (e.g., Meier et al. 2021; image adapted from NASA 2024, public domain). Additional abbreviations: NSIDC CDR, National Snow and Ice Data Center Climate Data Record; SAR, synthetic aperture radar.

and summer melt and breakup. The high-latitude polar oceans are especially challenging for field campaigns to access during polar winter, when ice is thick and storms are powerful. Visible satellite imagery is limited to months with good light, visibility, and cloud-free conditions. Biases in sea ice thickness observations from remote sensing, many of which relate to snow and surface properties, complicate our understanding of sea ice congelation growth. The recent development of models for the evolution of the sea ice FSD has further highlighted seasonal bias: While different methods and theories exist for the fracture of sea ice floes (although this too is far from settled), there is considerable uncertainty on how and to what extent fragmented sea ice heals during freeze-up. Existing schemes are ad hoc (Bennetts et al. 2017) or based only on geometrical considerations (when sea surface temperatures are below freezing) (Roach et al. 2018a, Boutin et al. 2020). The latter scheme has been tested with field observations (Roach et al. 2018b), but the generality of the welding rate and its sensitivities should be tested further. Field observations also suggest that an existing parameterization for lateral growth (Horvat & Tziperman 2015) may be too conservative (Roach et al. 2018b). These poorly constrained wintertime processes may have implications for sea ice evolution that impact other seasons.

Our process understanding of seasonal sea ice is notably biased toward the Arctic. Because it is more accessible to human activity, the Arctic is in general much better observed and understood

than the Antarctic; a Google Scholar search returns approximately 132,000 hits for “Arctic sea ice,” versus 33,000 hits for “Antarctic sea ice.” Both satellite sea ice observations (e.g., identification of the 80% sea ice concentration contour; Stroeve et al. 2016) and climate models (Roach et al. 2020) have greater skill in the Arctic than the Antarctic, and the Antarctic sees many fewer field campaigns than the Arctic. The Arctic bias is particularly evident for sea ice dynamics; most of the studies cited in Section 4.2 focus on the Arctic. The existing sea ice rheology models have been developed and tested for the compact central Arctic ice pack, with the focus on reproducing the characteristic features of ice deformation there (localization and multifractality of deformation, linear kinematic features, etc.). Similarly, satellite products of ice drift and deformation have been more widely tested against buoy and other in situ data in the Northern Hemisphere and so are likely more reliable there. Few studies consider ice deformation in the Antarctic based on ice drift products (an exception being Tian et al. 2022), and satellite coverage appears to be lower than in the Arctic (CMEMS 2024), although this is anticipated to improve with upcoming satellite missions. Coverage by autonomous platforms (e.g., ice mass balance buoys and ice-tethered profilers) is poor to nonexistent in the Antarctic.

Prior research has been biased toward certain research topics, arguably at the expense of others. For example, the past few decades have seen a strong focus on wave attenuation in sea ice cover. Despite this, the community has not come to a consensus on theory (e.g., Shen 2022, Thomson 2022). The disparate approaches to modeling wave attenuation are evident in the variation in attenuation rates (Rogers et al. 2016, Squire 2018) and subsequent impact on the area of sea ice impacted by waves in coupled wave–sea ice modeling (Cooper et al. 2022). Difficulties reconciling observations and models can be attributed to several factors, one of the most important being the fact that different types of information are obtained from the two. Observations, based mostly on satellite data or drifting buoys (e.g., Stopa et al. 2018, Kohout et al. 2020), provide information on the so-called apparent attenuation, i.e., the net change of energy of a given spectral component between two points. In most cases, energy spectra from only two locations are analyzed (or spectra from several buoys are analyzed in a pairwise manner), so that the exponential form of the apparent attenuation curves is assumed rather than observed (e.g., Kohout et al. 2020, Alberello et al. 2022). Moreover, the apparent attenuation (Section 4.2.2) is influenced by a long list of processes accompanying wave evolution in sea ice—not only sea ice–related scattering and dissipation (by several different mechanisms within and under the ice) but also wave growth by wind, whitecapping in open water patches, and nonlinear wave–wave interactions. Thus, disentangling from observations the processes directly related to ice—which is needed for validation of wave-in-ice models—is extremely challenging if not impossible (Rogers et al. 2016). An additional obstacle is the lack or incompleteness of the accompanying data on sea ice properties, including ice type, thickness, and floe size, and their spatiotemporal variability, which are necessary for attributing the observed attenuation to particular sea ice types and conditions.

In general, many fewer studies have focused on the impact of waves on sea ice than on the impact of sea ice on waves. We are beginning to see studies on wave-induced sea ice fracture and WRS (e.g., Boutin et al. 2020, Voermans et al. 2020), but further research is required. A key challenge is to come up with a model for wave-induced sea ice fracture that does not require the assumptions of total flexion or total rigidity but tends toward these extreme cases for short and long waves. Establishing a general theory for wave fracture will enable us to better characterize the resulting FSDs, interpret current and future observations, and parameterize this process for large-scale models in a computationally efficient way. As fracture can rapidly and extensively change the sea ice FSD, exposing the edges of floes to the ocean, it could play an important role in feedback loops (e.g., Asplin et al. 2014).

Many processes unrelated to waves in the SIZ are critical to seasonal evolution and deserve further investigation. In terms of thermodynamics, fundamental uncertainties remain in melt and growth rates and how these influence sea ice floe size and thickness. As discussed above, seasonal biases have led to uncertainties in growth rates through the winter. Yet even melt rates are subject to substantial uncertainties, although they have been relatively well studied. For example, observational constraints on lateral melt are notably sparse, likely due in part to the locations where it can be expected to dominate (e.g., the SIZ) and the three-dimensional nature of measurements (Richter-Menge et al. 2001). Surface melt is driven largely by the seasonal cycle of albedo, which has been observed to be relatively robust in the Arctic (Light et al. 2022) but has not been robustly quantified in the Antarctic, as the signal is likely dominated by the distribution of snow. In addition, basal growth rates are heavily impacted by the redistribution of snow across sea ice, as snow significantly impacts the vertical heat conduction through the ice and hence the capacity for congelation growth. In the Arctic, the redistribution of snow onto thin ice may impact winter growth rates by up to 8% (Clemens-Sewall et al. 2022), and large but uncertain losses to leads could have an even larger impact in the Antarctic (Leonard & Maksym 2011). Future work to observe heterogeneity in snow mass balance and to parameterize key processes is needed.

5.2. Connections Across Scales

In general, connecting the small-scale processes observed during field campaigns to their occurrence and generality throughout the sea ice cover remains difficult. The occurrence and seasonality of different ice types, which would hint at the processes that shaped them, are more or less unknown. Seasonal ice concentration budgets, computed from satellite observations of sea ice concentration and drift, offer information on where sea ice forms and melts at both poles (Holland & Kimura 2016). However, without sea ice thickness observations, such products only offer a two-dimensional view. A large-scale observational characterization of different types of sea ice growth and melt (e.g., a breakdown of sea ice surface, basal, and lateral melt) is lacking. This information would advance understanding of seasonal sea ice and would be especially helpful in validating large-scale climate and sea ice models, many of which now provide these diagnostics Keen et al. (2021).

Large-scale sea ice modeling for climate and Earth system models in the past has often taken place separately from research communities with a deep understanding of the relevant processes developed via observations and small-scale, physics-based modeling. As sea ice is tightly coupled to the ocean and atmosphere, it is strongly impacted by biases in either component, and sea ice parameters are often tuned to compensate for climate biases (e.g., Kay et al. 2022). This tuning makes it difficult to quantify to what extent higher sea ice process complexity improves simulation of sea ice and polar climate. The time taken between sea ice parameterization design and testing in a fully coupled model can be lengthy given model spin-up and ensemble size requirements; simulations without a free-running atmosphere and ocean are computationally cheaper but limit sea ice feedbacks. Simulations that constrain the large-scale atmospheric circulation while preserving coupled sea ice–ocean–atmosphere interactions in the boundary layer could be a promising intermediate path forward (e.g., Pithan et al. 2023). A key challenge for representing small-scale seasonal sea ice processes is to appropriately balance process complexity with model computational expense. In particular, the computational expense of current spectral wave models limits advances in modeling wave–ice interactions. As large-scale continuum sea ice models increase in spatial resolution and, in some cases, turn to multiresolution meshes, it will become increasingly important to explore how oceanic and atmospheric fluxes and dynamics interact with sea ice across

spatial scales. Besides horizontal resolution, increases in vertical ocean resolution might have an important influence on sea ice and upper-ocean coupling.

Discrete-element modeling (DEM) offers exciting paths forward for developing a better understanding of sea ice processes (e.g., Manucharyan & Montemuro 2022, Åström et al. 2023, Brenner et al. 2023). However, it is key that these efforts are integrated with large-scale modeling approaches to ensure maximal research advances across spatial scales. Future large-scale sea ice models may have superparameterizations embedded within them or include parameterization informed by DEM. To help bridge the gap, it may be possible to combine discrete-element and continuum models, e.g., by resolving large floes with a discrete-element model and small floes or grease ice with a continuum model (Marquart et al. 2023). Machine learning has potential to support existing models by replacing expensive parameterizations (Horvat & Roach 2022) or correcting forecasts (Finn et al. 2023). It will be increasingly important to bring together climate modelers and process experts to ensure that new modeling technology—whether related to resolution, noncontinuum models, or machine learning—improves process fidelity.

5.3. Future Change

SIZs in both Arctic and Antarctic regions are rapidly changing. The Arctic has been losing multiyear ice for several decades (e.g., Kacimi & Kwok 2022). In 2023 and the early months of 2024, Antarctic sea ice has been anomalously low, with perennial sea ice remaining only in the Weddell Sea (Meier et al. 2021). Wave heights are increasing in both hemispheres as winds intensify (Liu et al. 2024). In the future, a mostly seasonal ice pack may respond differently to atmosphere and ocean forcings, resulting in different feedbacks between thermodynamic and dynamic processes. Some key questions that address future changes in the coupled sea ice system include the following:

- Will the seasonal cycle of sea ice albedo change with a thinner and more seasonal sea ice pack? Anticipated trends toward thinner ice, lower snowpack, and higher melt pond fractions in both hemispheres would result in lower average albedos. Might this reduce the efficiency of the sea ice albedo feedback?
- How will the partitioning between top, lateral, and basal melt change? Basal melt is likely to increase due to anticipated changes in ocean heat content (e.g., Cheng et al. 2022). Lateral melt may also increase from increasingly dynamic forcings and shifts in FSD in both hemispheres. Might this increase the efficiency of the sea ice albedo feedback?
- How will sea ice drift change? As sea surface height increases, summer Arctic sea ice drift may slow down (Ward & Tandon 2023). Will sea ice be less compact on average, and will this impact our understanding of sea ice dynamics?
- As ocean surface waves increase, what role will they play in sea ice feedbacks? In both hemispheres, increasing wave heights might make fragmentation more important and promote the formation of pancake ice, which can be a more effective method of sea ice formation. What will the net impact on feedbacks be?
- The connections between sea ice growth and melt and upper-ocean heat, salinity, and stratification are understood to be important in the Antarctic SIZ (e.g., Wilson et al. 2019) but are poorly observed. They may become more important in the Arctic SIZ with Atlantification under warming (e.g., Asbjørnsen et al. 2020). How will these connections determine future SIZ variability?

As the planet continues to warm, different seasonal sea ice processes have the potential to play an important role in both negative and positive feedback loops. These feedback loops will have

implications for local and global climate as well direct impacts on hunting and fishing, ecosystems, coastal erosion, and native communities. Our community will make the most progress by addressing interfaces between subdomains: reconciling observations and modeling, improving process understanding across spatial scales, contrasting the Arctic and Antarctic, and considering the boundaries between different components of the coupled sea ice–ocean–atmosphere system.

DISCLOSURE STATEMENT

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